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Dear Hiring Manager,

I am writing to apply for the AI Software Developer role at Stingray. My research is built almost entirely on music-content analysis: I designed and published MIRaaS – Music Information Retrieval as a Service – a UDP edge-to-server offload architecture for MIR workloads, with a latency-characterization paper accepted at IEEE IS2 2026. Alongside that, I created and released four open music-content datasets on Zenodo (DoMP, DoPP, DoDP, DoDP2), totalling 9,800+ recordings from 70+ musicians. Between the two, I bring direct, concrete experience with the kind of content-analysis and catalog infrastructure a music and media company depends on.

That research sits on top of an engineering track record in real-time audio ML: my published pattern detector runs in 14 ms on a Raspberry Pi 4 at F1 0.76, 74x faster than the DTW baseline it replaced, and I maintain Nebula, a public C++17 audio-feature-extraction and MIR toolkit with per-feature ARM latency benchmarks. I also bring 15+ years as a performing and session guitarist, which gives me musical-domain intuition that I have found genuinely changes how I approach audio-content problems, not just the engineering behind them. I already have ties to Montreal from a 2024 visiting research appointment at McGill University's IDMIL/CIRMMT lab, so the city and its music-tech community are familiar ground.

The posting is light on day-to-day scope, so I do not yet know exactly which parts of Stingray's product and catalog stack this role touches. That is not a concern on my end – the domain fit is exactly where I want to be – and I would be glad to learn the specific business context and priorities quickly once I understand where the team needs the most help.

I am a Canadian Permanent Resident and do not require sponsorship to work in Canada. I am available immediately and open to relocating to Montreal, a city I already know through my McGill collaboration. I would welcome the opportunity to discuss how my music-technology research and real-time ML engineering background could contribute to Stingray's work.

Sincerely,
Nishal Stanislaus Silva